

1. Writing Data To a File

```
#include<iostream>
#include<fstream>
using namespace std;

int main()
{
    char ch = 'x';
    int j = 77;
    double d = 6.02;
    char str1[] = "Pakistan";
    char str2[] = "Islamabad";

    ofstream outfile("output.txt");

    outfile << ch << endl << j << endl << d << endl << str1 << endl << str2 << endl;
    cout << ch << endl << j << endl << d << endl << str1 << endl << str2 << endl;

    outfile.close();

    return 0;
}
```

2. Reading Data From a File

```
#include<iostream>
#include<fstream>
using namespace std;

int main()
{
    char ch;
    int j;
    double d;
    char str1[80];
    char str2[80];

    ifstream infile("output.txt");

    infile >> ch >> j >> d >> str1 >> str2;
    cout << ch << endl << j << endl << d << endl << str1 << endl << str2 << endl;

    infile.close();

    return 0;
}
```

3. Strings with Embedded Blanks

```
#include <fstream>
using namespace std;

int main()
{
    ofstream outfile("test.txt", ios::out);
```

```

outfile << "I fear thee, ancient Mariner!\n";
outfile << "I fear thy skinny hand\n";
outfile << "And thou art long, and lank, and brown, \n";
outfile << "As is the ribbed sea sand.\n";

outfile.close();

return 0;
}

```

4. Extracting from file

```

#include <fstream>
#include <iostream>
using namespace std;

int main()
{
    const int MAX = 80;           //size of buffer
    char buffer[MAX];             //character buffer

    ifstream infile("test.txt", ios::in); //open a file for reading

    //check if file opened successfully
    if (!infile)
    {
        cout << "Error opening file!" << endl;
        return 0;
    }

    //read and display lines from the file
    while (!infile.eof())          //until end-of-file
    {
        infile.getline(buffer, MAX); //read a line of text
        cout << buffer << endl;      //displaying it
    }

    infile.close();                //close the file

    return 0;
}

```

5. Character I/O

```

#include <fstream>
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string str = "Time is a great teacher, but unfortunately it kills all its
pupils. Berlioz\n";

    ofstream outfile;
    outfile.open("test2.txt", ios::out);
}

```

```

//check if file opened successfully
if (!outfile)
{
    cout << "Error opening file!" << endl;
    return 0;
}

for (int j = 0; j < str.size(); j++)
{
    outfile.put(str[j]);           // similar to outfile << str[j];
}

outfile.close();
cout << "File written\n";

return 0;
}

```

6. Reading this file back and displaying

```

#include <fstream>
#include <iostream>
using namespace std;

int main()
{
    const int MAX = 80;
    char buffer[MAX];

    ifstream infile;
    infile.open("test2.txt", ios::in);

    //check if file opened successfully
    if (!infile)
    {
        cout << "Error opening file!" << endl;
        return 0;
    }

    while (!infile.eof())           //until end-of-file
    {
        infile.getline(buffer, MAX); //read a line of text
        cout << buffer << endl;     //display it
    }

    infile.close();                 //close the file

    return 0;
}

```

7. Appending to a File

```

#include <iostream>
#include <fstream>
using namespace std;

int main()
{

```

```
ofstream outfile;
outfile.open("test2.txt", ios::app); // Open file in append mode

if (!outfile)
{
    cout << "Error opening file!" << endl;
    return 0;
}

outfile << "This line is appended to the file." << endl;
outfile << "Appending another line..." << endl;

outfile.close();

cout << "Data appended successfully!" << endl;

return 0;
}
```